

1 Structure of the DLG Format

Basics of HYDRA BAPI

Data is always posted to the database in accordance with basic guidelines ensuring their consistency and uniformity. This is why any writing access to the database is performed by programs providing a uniform interface to this end.

This means that all writing accesses to the HYDRA database irrespective of whether these are called via HYDRA applications or external applications/ systems are executed by a program with a defined interface.

This is mainly the HYDRA BAPI. It is used in the course of the master data transfer in order to transfer and post data provided by and processed in external systems to HYDRA.

BAPIs and dialog commands

Essentially, there is for each object (that can be maintained using the MOC) such a BAPI in HYDRA. Objects in this sense may be (production) orders or master data records. There are always different methods to access such an object. In the easiest case this is a method to create (INSERT), modify (UPDATE) or delete data records (DELETE).

In more complex cases and/or when this is requested by the application, also different methods are implemented. This may be modifying methods comprising an insertion or modification or additional application-specific methods.

Such a BAPI is called by a so called dialog command. This command is comprised of:

<Object>.<Method>

This is an exemplary (and incomplete) overview of the available objects and their selected methods

Object	Methods	Comment
ANR	INSERT UPDATE DELETE MODIFY	The ANR object designates the order.

Object	Methods	Comment
MNR	INSERT UPDATE DELETE	The MNR object designates machines/ workplaces.
FERTVAR	INSERT UPDATE DELETE	The FERTVAR object designates production variants.
RES	INSERT UPDATE DELETE	The RES object designates the resources of the module WRM and DNC.

Dialog data strings

After the initial BAPI call using the command, the use data will be transferred in a so-called dialog string or dialog data string. The use data in a dialog string are clearly identified by indicators, also designated as acronyms.

Such an acronym may represent at least one database field or also have controlling effects on postings. The acronym is always followed by the equal sign "=" and the value transferred for this acronym. The individual acronyms and their values are separated by pipes "|" from each other and from the dialog command.

Example:

```
DLG=FERTVAR.INSERT|FERTVAR.ATK=BLOO01052225000000|FERTVAR.MGRP=BW2000|
FERTVAR.RESTYP=WNR|FERTVAR.RES=BLOO01052225000000 2|
FERTVAR.SZY=17143|FERTVAR.TLG=2|
FERTVAR.BEM=BLOO01052225000000\|rose\|2\|rose\|BW2000|
FERTVAR.VER=1|FERTVAR.STA=F|FERTVAR.FIR:ATK=0|
```

Data formats/ mandatory acronyms

The descriptions of the acronyms are based on the following data types:

Type	Description
CHAR x	For the data type CHAR the information will be aligned to the left; unnecessary positions will be filled with blanks. Example: "ABCD "
NUM x	Numeric field of the length x without sign. For the NUMC data type only digits are allowed (ASCII-digits 30 hex to 39 hex). The numbers will be aligned to the right and unnecessary positions will be filled with zeros. Example: "00000002"
DEC x.y	Numeric field of the length x contains y decimal places. A data field in the HYDRA format is preceded by a sign ("+" or "-") and it contains a decimal point. Empty places must be filled with zeros. e.g. DEC 13.3: -1234567890.123

Each BAPI call **must** contain the following header data in the dialog data

Identification	Content	Description
DLG	{BAPI call}	Dialog identification: This dialog identification indicates the desired BAPI call
USR	NUM 4	HYDRA user: This Hydra user number uniquely identifies a HYDRA client:
		MOC: USR = 20000 + MOC number USR = 20000 + MOC
		LAN terminal (LANT) FB terminal (FBT): USR = 2000 + terminal number USR = 2000 + TNR
		External terminals USR = 3000 ... 3999
		MLE-MDM USR=9999
DAT	{mm/dd/yyyy}	Date: current date in the format mm/dd/yyyy "Today" can be used as placeholder for the dynamic determination.
ZEI	{seconds}	Time: current time in the seconds format "Now" can be used as placeholder for the dynamic determination.

Depending on the BAPI call, additional identifications must/ may be entered.

Data objects with files

Only the file names will be indicated in the dialog string for such objects that contain files in addition to the data fields of dialog data strings, e.g. document resources or DNC resources. The files themselves will be stored to defined data areas. The data import consists of two steps: Dialog data strings and files.

Dialog data strings - acronyms

The acronym to indicate the file is a field of the field type CHAR 128 that includes the file name. In most of the cases the name is only indicated without path - please see the documentation for the BAPI concerned.

Example: RES.SPEICHORT:DATA includes the file name without path and without extension of the attached DNC file. The storage location and the extension are defined before in the system via the resource type.

File format:

The file format is not important for the storage in HYDRA. The file will be stored to the specified storage location. The application will then interpret this file. For the import of master data it must be taken into account that the file must be stored to the directory specified for the application.

Example DNC files: The DNC type defines in which folder the files are and how they must be stored and interpreted.

Multilingual database contents

As part of SIS-HLM, there is now the possibility to define descriptive texts in several languages for specific objects in the database. Provided that this function is enabled on the system, these columns may generally also be filled by using the master data import. Please note the following:

- Specify the target language

The target language can be transferred as additional acronym in the dialog data strings.

Example:

Machine master data is to be transferred. English (EN) is defined with language index 2 in the system. The dialog data string to transfer this data has to be structured as follows:

DLG=MNR.INSERT|...|MNR.MNR=<Machine>|MNR.BEZK=English description|...|LANG=2|...

- Only one language can be transferred every time an import is started.

This means, that two or more import runs might be required, subject to the number of configured languages. Please note the following:

- The first import has to be performed using the *.INSERT method.
This rule can be ignored if there is a method "/*.MODIFY" for the object. As in this case, the system decides whether an INSERT or an UPDATE is to be performed.
- All other imports need to be performed by way of the method "/*.UPDATE" indicating all key fields pertaining to the object, the language-dependent description and the target language using the acronym "LANG=n".
- If the system uses a separately generated, internal key for an object, this one has to be determined after the initial creation. This internal key then needs to be provided for the updates that follow.